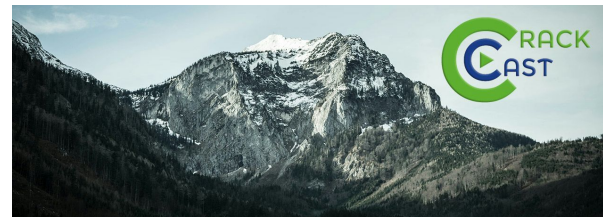


Decision to tube

1. Failure to maintain or protect the airway
 - Can't handle gastric and oral secretions
2. Failure to ventilate or oxygenate
 - ABG rarely useful
3. Pts' anticipated clinical course and likelihood of deterioration
 - multiple trauma, open#, neck trauma, burns

Identifying the difficult airway (to plan steps 1-2-3-):

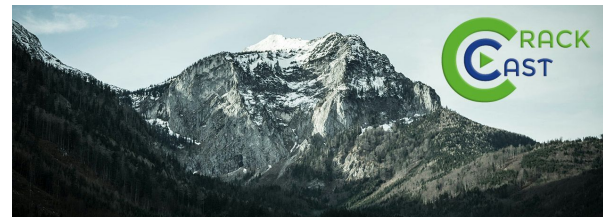
- Diff. DL
 - "The lemon test" if yes to anything: tread carefully!
 - LEMON
 - look for gestalt signs :
 - ***KEY sign is the presence of an underbite: if the patient CANNOT bite their upper lip with their lower teeth (buck teeth boy) they will be a very difficult tube!***
 - evaluate the 3-3-2 rule
 - A high riding larynx and receding mandible = impossible tube
 - Mallampati score :
 - use a Laryngoscope to visualize in obtunded pts.
 - Obstruction or obesity
 - epiglottitis, ludwig's angina,
 - Neck mobility
 - Ank. spondylosis, Rheumatoid Arth,
- Diff BVM
 - MOANS
 - Mask seal
 - Obstruction or obesity
 - Age > 55
 - No teeth
 - Stiffness of chest
 - asthma, COPD, pregnancy
- Diff. Extraglottic device
 - RODS
 - Restricted mouth opening
 - Obstruction/obesity
 - Distorted anatomy
 - Stiffness to ventilate
- Difficult Cricothyrotomy
 - SMART
 - Surgery
 - Mass (abscess, hematoma)



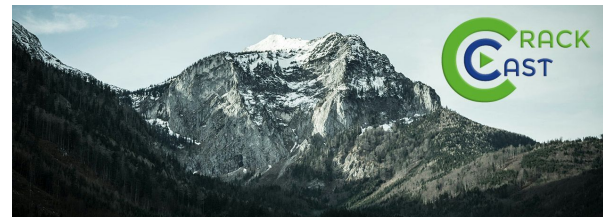
- Access/anatomy problems (obesity, edema)
- Radiation
- Tumor
- Cormack Lehane grading view
 - 1-4
 - 4 = not even the epiglottis seen.
 - 2a = arytenoids and portion of vocal chords, 2b=arytenoids only
 - POGO system can also be used
- Confirmation of placement
 - most serious complication: unrecognized esophageal tube: and anoxic injury**
 - Traditional methods (misting, listening over lungs, chest rise) are unreliable
 - Colorimetric ETco2
 - Yellow = yES!
 - ***need to do at LEAST 6 manual breaths to confirm placement
 - in cardiac arrest c02 detection will be impaired
 - Ultrasound of trachea
 - palpation during intubation
 - Aspiration technique (With cuff deflated)
 - Quantitative colorimetric ETco2
 - A single chest radiograph **doesn't** confirm placement (AP overshadows esophagus)
 - Causes of failure of ETco2
 - complete obstruction
 - asthma

Approach to intubation:

- **Questions to ask:**
 - ***is this a crash airway?***
 - ***Is this a difficult airway?***
 - ***no to both = do RSI***
- Crash airway = no drugs, except a single LARGE dose of Succx.(2mg/kg) if unable to tube
- Failed airway =
 - failure to intubate despite 3 attempts at RSI
 - i. ****failure to oxygenate using BVM or EGD****
 - OR: experienced clinician identifies a grade 4 view on their first attempt.
- Difficult airway
 - is predicted based on the previous mnemonics
 - ***NMBAs should not be used unless:
 - i. you are forced to act
 - ii. intubation is likely to be successful
 - iii. oxygenation via BVM or EGD is possible



- If a difficult airway is predicted and oxygenation is OK
 - i. RSI, with double setup and backup ready
 - ii. Awake intubation
- Failed airway
 - IF oxygenation cannot be maintained = failed airway --> Cric!!
 - i. unless you have an EGD immediately available
 - ONLY one attempt should be made with this and it shouldn't delay cric!
- Methods of intubation
 - RSI = "simultaneous administration of a potent sedative and NMBA"
 - i. No interposed assisted ventilation
 - ii. Requires a period of preoxygenation (3 mins) and the other 7 P's
 - iii. Desat times (<90%)
 - Adult healthy: 8 mins
 - adult, sick: 5 mins
 - obese adult: 2-3 mins
 - child: 3-4 mins
 - iv. Pretreatment (at least 3 mins bf. induction)
 - For three at risk groups:
 - a. Reactive airways: +/- Lido 1.5 mg/kg
 - b. CV disease: Fentanyl 3 mcg/kg/*
 - c. Inc. ICP: +/- Lido 1.5 mg/kg + Fentanyl 3 mcg/kg
 - v. Postintubation mgmt!
 - Blind nasotracheal intubation
 - Awake oral intub.
 - i. Sedative and topical agents
 - Oral intubation without pharm agents
 - i. eg. crash airway
- Pharm agents:
 - NMBA's
 - i. do not cross placenta or BBB
 - ii. work by mimicking the Ach molecule
 - Succinylcholine : Depolarizes the motor end plate at Ach
 - a. decreased plasma pseudocholinesterase increases the amount of Succ. available
 - b. 45 sec to effect.
 - c. spont. respiration recover at 6-10 mins
 - d. 1.5 mg/kg regardless of obesity
 - minor cardiovasc effects (bradyC. in some kids)
 - fasciculations and myalgias
 - hyperkalemia



- ! up to six months
post injury (stroke), never with MS, ALS, muscular dystrophy
- careful in renal failure patients
 - masseter spasm: known occurrence
 - trxt: rapid bolus of Rocuronium and tube em!
 - Malignant Hyperthermia
 - never been reported after an ED intubation
 - requires refrigeration
- Rocuronium: nondepolarizing, competitive (prevents access of Ach on the receptors)
 - a. dose: 1-2 mg/kg
 - b. lasts 50 mins
 - c. onset 60 secs
 - d. NO contraindications
- Induction agents:
 - i. Etomidate
 - hemodynamically stable. 0.3mg/kg IV
 - a. decreases ICP and cerebral blood flow without affecting MAP
 - b. brief myoclonus
 - c. prolonged doses causing adrenal suppression **not clinically relevant
 - ii. Ketamine
 - 1-2 mg/kg. IV. onset in 30 sec, peaks in 1 min, 10-15 mins duration
 - creates a cataleptic state
 - excellent choice in unstable patients, asthmatics, sepsis, trauma
 - causes catecholamine release, bronchodilation
 - NO clinically relevant ICP, CBF_{low},
 - a. *maybe* avoid in the hypertensive TBI *
 - can cause emergence reactions (Trxt with benzos)
 -
 - iii. Propofol
 - 1.5-2 mg/kg IV
 - decreases ICP, cerebral O₂ useage, myocardial depression
 - hypotension, pain at site of administration
 - BE cautious: elderly, hemodynamic compromise/cardiac instability,
 - Not be used in those with a soybean/lecithin/egg protein anaphylactic allergy



iv. Midazolam

- 0.2-0.3 mg/IV
- onset 30 sec. duration 15-20 mins.
- cautious reduced doses in the elderly.

v. Intranasal sedation:

- Intranasal sedation protocol: (from Intranasal.net)

General points:

- Midazolam, ketamine, dexmedetomidine and sufentanil are the most commonly used sedative medications for IN delivery.
- Midazolam results in mild somnolence with resultant reduction in anxiety and probably amnesia. It will not make the patient unconscious.

Be aware that midazolam causes some nasal burning for 30-45 seconds when administered.

- *In small children you should administer lidocaine 2% or 4% - 0.2 ml per nostril 5 minutes prior to the midazolam to stop the burning.*

Combination therapy with midazolam plus either sufentanil or ketamine may work better than any of the medication alone. Reasonable IN starting dose:

- **Midazolam 0.4 to 0.5 mg/kg**

Use the lower dose for minor, non-painful procedures such as radiographic imaging

Use the higher dose for better sedation prior to procedures such as laceration repair

Ketamine 10 mg/kg

Midazolam plus ketamine: 0.2 to 0.3 mg/kg of midazolam plus 5 mg/kg of ketamine

Use only **concentrated midazolam (5 mg/ml) and ketamine formulations**

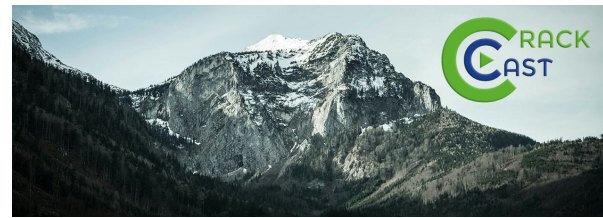
Be sure to monitor oxygen saturation in all patients

Ideal volume is 0.3 to 0.5 ml per nostril, maximum is 1 ml per nostril, more will just run out nose.

Nasal naloxone and flumazenil can be used as reversal agents

SPECIAL INTUBATING CIRCUMSTANCES

- Status Asthmaticus
 - **unsuccessful RSI followed by prompt cricothyrotomy (No EGD or BVM!)**
 - Pre-treat with lidocaine 1.5 mg/kg IV
 - Beta agonist in line - continuous albuterol nebs
 - Ketamine
- Reflex sympathetic response to laryngoscopy
 - Important to mitigate this in
 - i. ***Increase ICP
 - ii. ICH / SAH / Aortic dissection / Ischemic heart disease
 - iii. Fentanyl 3 mcg/kg IV over 60 seconds** (prevents rigid chest, apnea,)
- Elevated ICP
 - need to reduce secondary neuro. injury by keeping MAP at 100
 - treatment



- i. elevated HOB
 - ii. pretreat with fentanyl
3 mcg/kg IV over 60 sec
 - iii. IF time permits also pretrxt with Lidocaine 1.5mg/kg
 - iv. Etomidate / Ketamine + NMBA
- C Spine injury
 - VideoLyrngsp
 - i. less lifting force or Spine mvmt
 - critically think if a cspine injury is actually suspected-based on mechanism, neuro exam
- Pediatric tube
 - Airway differences:****
 - i. higher larynx (more acute angle oropharynx to larynx -- may need BURP)
 - ii. high, soft and floppy epiglottis
 - iii. prominent occiput (put towel under shoulders)
 - iv. short airway (dont tube the bronchus)
 - need a higher dose of Succx. (2mg/kg)
 - may need jet ventilation in young children ?<8-10yrs

airway devices

- DL
- VL
 - for technique see EMCRIIT podcast: put tube in mouth before blade if narrow opening
 - introduce: middle of mouth, rigid stylet (lube it up first),
- Intubating stylets/scopes
 - useful in angioedema, blunt neck trauma, awake intubation, burn patients,
- EGD's
 - LMA, ILMA, I-gel
 - King, combitube---designed to have you intubate the esophagus and then ventilate through a side port
- surgical mgmt
 - needle cricothyrotomy with transtrach. jet ventilation
 - kids with supraglottic obstruction
 - cricothyrotomy
 - **used in a can't intubate , can't ventilate scenario**
 - scalpel, finger, bougie and 6.5 cuffed tube.



- Case 1
 - Obese male: with pneumonia/pneumosepsis
 - Preparation:
 - prepare patient, equipment, drugs, mentally go through your plan with the team!!! and the room!!!
 - rehearse the recipe with the room
 - rehearse what will happen if it goes south
 - epi, atropine, etc.
 - surgical airway
 - someone tells me when the sat is 94...
 - preoxygenation
 - technically 3 mins or 8 TV breaths
 - if no success with NRB then: (SP02<95%)
 - think about shunt physiology: (ARDS, atelectasis, etc)
 - BVM with a PEEP valve or CPAP/BIPAP as a temporizing measure
 - using the BVM: breathe WITH the patient
 - hold the BVM on the face the entire cycle and squeeze the bag with inspiration, and hold the mask in place for expiration
 - if no effect prepare for intubation
 - Positioning
 - ear to sternal notch
 - "face parallel to the ceiling"
 - "ears level with the sternal notch"
- Case 2:
 - 21 year old asthmatic, with no effect with standard asthma treatment:
 -
 - DSI!!! PAPER as evidence
 - Ketamine 1 mg/kg and 0.5mg/kg prn for
 - BIPAP/CPAP and or MASK when calm
 - q15 min ABG's, or VBG's (PCO2 and Ph)
 - Ketamine infusion
 - "I recognize that a RELATIVE contraindication to NIPPV" is altered LOC
 - but based on recent studies it has been shown to be safe
 - This is a known difficult airway in a forced to act scenario
 - Paralytic agents
 - sugammadex reversal agent for Roc is coming down the pipeline
 - the decision to intubate is key - no matter what your drug choice, if you choose to paralyse you have to bear with the repercussions
 - I want the BEST chance on my first attempt.
 - Roc.
 - in every case except status epilepticus



- buys time to get postintubation care...lines , tuuuubes, meds
- Succ.
 - can't redose it!!!
 - slow respiratory return.....
 - shorter time to desaturation!!!

Case 3:

- multitrauma case with bad airway
- when and how do you teach
- Videolaryngoscopy view:
 - turn the tube COUNTER CLOCKWISE if it's hitting the cords

Case 4:

- obese, no neck, aspiration, needs intubation:
- DON'T paralyse
 - try and awake look. !!!
 - think about putting a guidewire in the neck and then look. from above that way if you need to cut you know your anatomy.

Case 5 exchanging a tube:

- ASSUME that it's a brand new intubation, have everything ready with paralytics and meds and new equipment just in case
- don't half ass it and assume it will go smoothly
- Angioedema case
 - if forced to act, consider an AWAKE CRIC
 - call for help early
 - nearly stupid to try from above! you only have ONE CHANCE
- GI bleed case:
 - if you oesophageal intubate leave the tube in, then you know where the tube is,
 - then try to intubate